

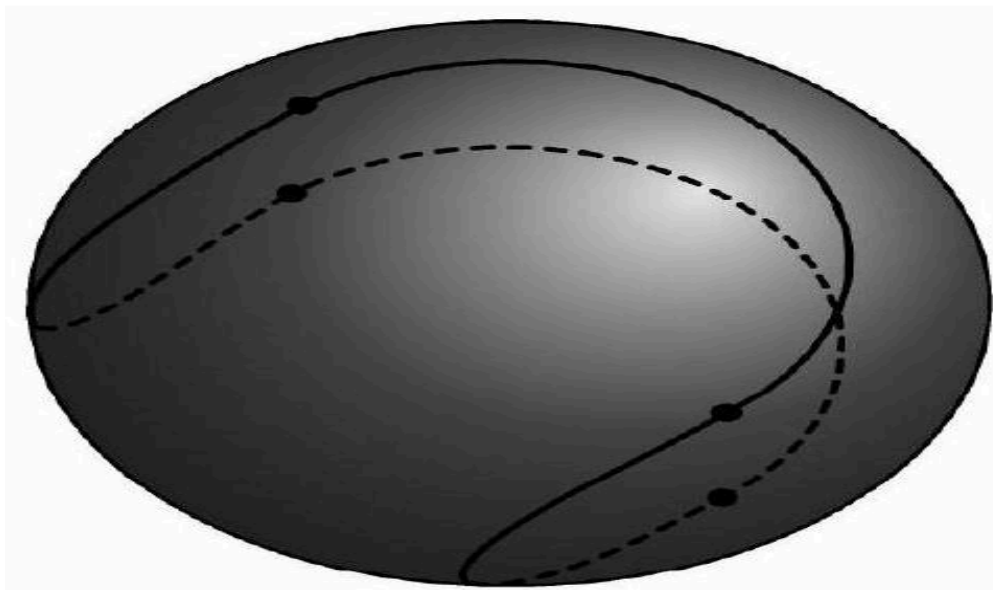
2023 AMC 10B Problem 20 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10B Problem 20. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10B Problem 20

Problem:

Four congruent semicircles are drawn on the surface of a sphere with radius 2, as shown, creating a closed curve that divides its surface into two congruent regions. The length of the curve is $\pi\sqrt{n}$. What is n ?



Answer Choices:

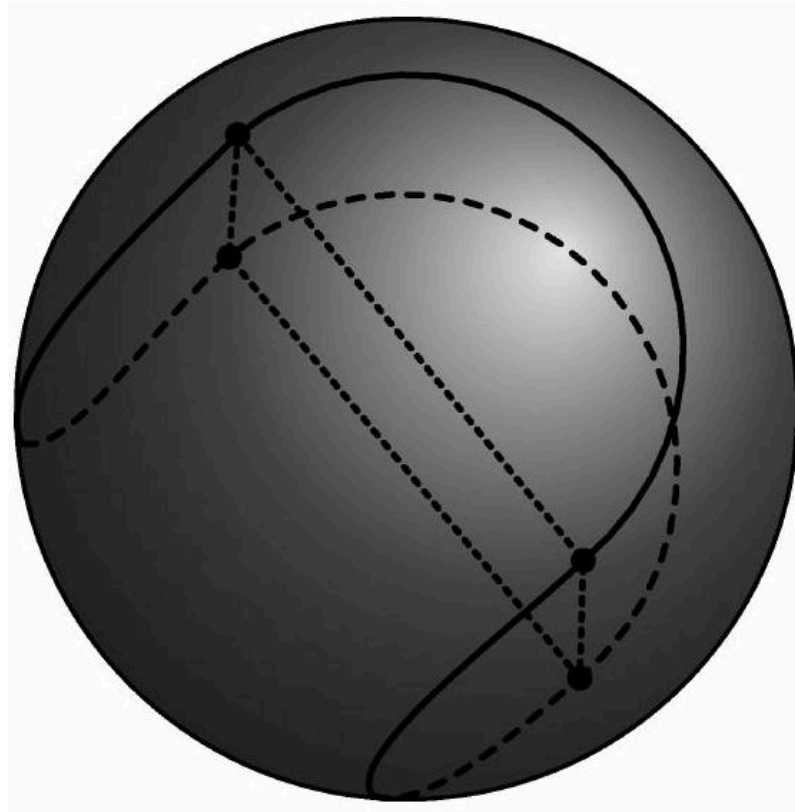
- A. 32
- B. 12
- C. 48

D. 36

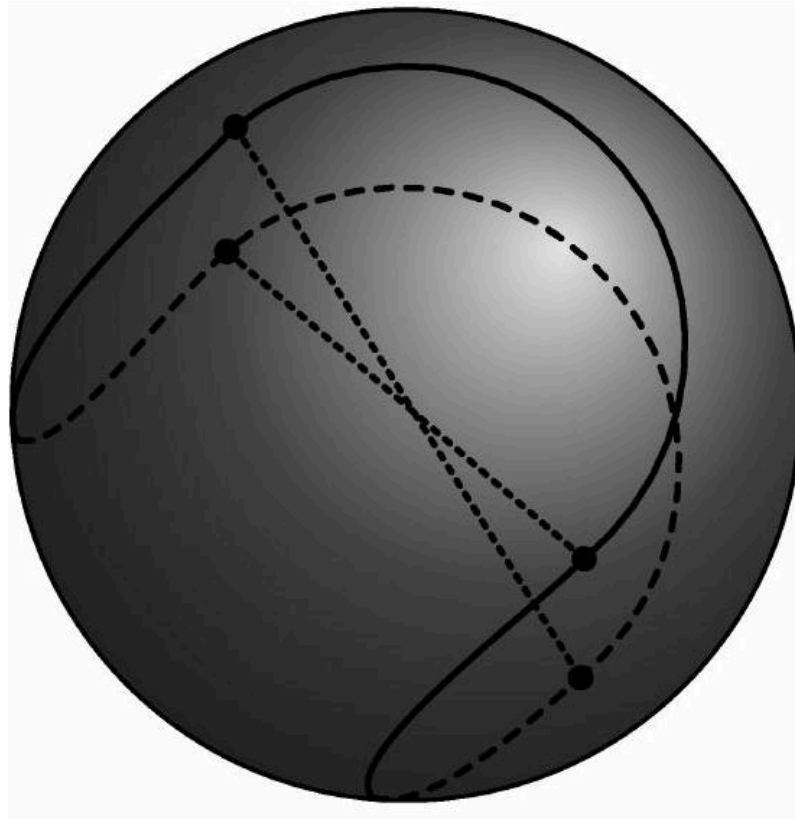
E. 27

Solution:

The four endpoints of the semicircles determine a square, as shown in the picture.



The diagonals of the square are diameters of the sphere, as shown in the next picture.



Each side of the square is a diameter of a semicircle, four of which comprise the closed curve. The diameter of the sphere is 4, so the diameter of the semicircles (the side length of the square) is $\frac{4}{\sqrt{2}} = 2\sqrt{2}$. Each semicircle has arc length $\frac{\pi d}{2} = \frac{\pi 2\sqrt{2}}{2} = \pi\sqrt{2}$, so the length of the curve is $4\pi\sqrt{2} = \pi\sqrt{32}$, and $n = 32$.

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